

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		C (1) contains two shells not three / less shielding of outer electron (1) of the elements with two shells it contains greatest nuclear charge / most protons (1)	2		1	3		
		(ii)		G (1) nucleus is most positively charged / has most protons (so greatest attraction to last electron) (1)	1		1	2		
		(iii)		D, E, G award (2) for all three correct award (1) for two correct / two correct and one incorrect answer award (1) for all three correct and one incorrect answer award (0) if more than one incorrect answer	1	1		2		
		(iv)		D, (E), G	1			1		
	(b)	(i)		(gaseous magnesium) atoms bombarded by electrons	1			1		
		(ii)		deflected through a magnetic field	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		^{28}Al		1		1		
		(ii)		21 hours		1		1	1	
	(d)			bonding consists of (a regular array of) metal ions surrounded by a 'sea' of delocalised / valence electrons (1) (electrostatic) attraction between ions and electrons (1) (when a force is applied) the layers of metal ions slide over each other forming a new shape (1) credit possible for appropriate diagram(s)	3			3		
	(e)			6.17×10^{-4}		1		1	1	
				Question 9 total	10	4	2	16	2	0